Fundamentals of Engineering Mathematics I

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Class 9; Series solutions of ODEs 1

Power series method

Convergence of power series

Legendre's equation

We are looking for the solution(s) of the ODE having variable coefficients.
There are two methods

- Power series method and
- Frobenius method

Power series method

We are looking for the solution in a form of power series.

A power series function expanded at a point x_0 is given or defined by

$$f(x)=\sum_{m=0}^{\infty}a_m(x-x_0)^m=a_0+a_1(x-x_0)+a_2(x-x_0)^2+\cdots$$

with

- non-negative integer *m*
- constant coefficients a_m
- center of the series x_0.

Ex)

$$\frac{1}{1-x}=\sum_{m=0}^{\infty}x^m=1+x+x^2+\cdots$$

$$e^x=\sum_{m=0}^{\infty}\frac{x^m}{m!}=1+x+\frac{x^2}{2}+\frac{x^3}{6}+\cdots$$

Further,

$$e^{i\theta} = \sum_{m=0}^{\infty} \frac{(i\theta)^m}{m!} = 1 + (i\theta) + \frac{(i\theta)^2}{2} + \frac{(i\theta)^3}{6} + \cdots$$

$$= 1 + \frac{(i\theta)^2}{2!} + \frac{(i\theta)^4}{4!} + (i\theta) + \frac{(i\theta)^3}{3!} + \frac{(i\theta)^5}{5!} + \cdots$$

$$= \left\{1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \cdots\right\} + i\left\{\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \cdots\right\} = \cos\theta + i\sin\theta$$

Ex)

For an ODE of

$$y' - y = 0 \quad \text{or} \quad y' = y$$

we set a power series function as a trial solution of the ODE as

$$y = \sum_{m=0}^{\infty} a_m x^m = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \cdots$$

Then, its derivative becomes

$$y' = a_1 + 2a_2 x + 3a_3 x^2 + \cdots$$

Substituting these functions into the ODE gives

$$y' - y = 0 \quad \Rightarrow \quad a_1 + 2a_2 x + 3a_3 x^2 + \cdots - a_0 - a_1 x - a_2 x^2 - a_3 x^3 + \cdots = 0$$

Collecting the terms of a like power of x gives

$$(a_1 - a_0) + (2a_2 - a_1)x + (3a_3 - a_2)x^2 + (4a_4 - a_3)x^3 + \cdots = 0$$

Because the power functions are linearly independent to each other, every coefficient should vanish, giving

$$a_1 - a_0 = 0 \quad \Rightarrow \quad a_1 = a_0$$

$$2a_2 - a_1 = 0 \quad \Rightarrow \quad 2a_2 = a_1 \quad \Rightarrow \quad a_2 = \frac{a_1}{2} = \frac{a_0}{2!}$$

$$3a_3 - a_2 = 0 \quad \Rightarrow \quad 3a_3 = a_2 \quad \Rightarrow \quad a_3 = \frac{a_2}{3} = \frac{1}{3}\frac{a_0}{2!} = \frac{a_0}{3!}$$

$$4a_4 - a_3 = 0 \quad \Rightarrow \quad 4a_4 = a_3 \quad \Rightarrow \quad a_4 = \frac{a_3}{4} = \frac{1}{4}\frac{a_0}{3!} = \frac{a_0}{4!}$$

Thus, by replacing, the solution is obtained simply as

$$y = \sum_{m=0}^{\infty} a_m x^m = a_0 + a_0 x + \frac{a_0}{2!}x^2 + \frac{a_0}{3!}x^3 + \frac{a_0}{4!}x^4 + \cdots = a_0 e^x$$

It is an exponential function!

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Ex) Legendre equation in spherical symmetry

For an ODE of (a Legendre equation in spherical symmetry)

$$\left(1-x^2\right)y''-2xy+2y=0$$

we set the trial solution as

$$y=\sum_{m=0}^{\infty}a_m x^m=a_0+a_1x+a_2x^2+a_3x^3+a_4x^4+a_5x^5+a_6x^6+\cdots$$

Then, its derivatives are calculated as

$$y'=a_1+2a_2x+3a_3x^2+4a_4x^3+5a_5x^4+6a_6x^5+\cdots$$

$$y''=2a_2+3\cdot 2a_3x+4\cdot 3a_4x^2+5\cdot 4a_5x^3+6\cdot 5a_6x^4+\cdots$$

Substituting these into the ODE, we have

$$y''=2a_2+3\cdot 2a_3x+4\cdot 3a_4x^2+5\cdot 4a_5x^3+6\cdot 5a_6x^4+\cdots$$

$$-x^2y''=-2a_2x^2-3\cdot 2a_3x^3-4\cdot 3a_4x^4-5\cdot 4a_5x^5+\cdots$$

$$-2xy'=-2a_1x-2\cdot 2a_2x^2-2\cdot 3a_3x^3-2\cdot 4a_4x^4-2\cdot 5a_5x^5+\cdots$$

$$2y=2a_0+2a_1x+2a_2x^2+2a_3x^3+2a_4x^4+2a_5x^5+\cdots$$

Collecting the terms of a like power gives

$$2a_2+0+0+2a_0=0 \quad\Rightarrow\quad a_2=-a_0$$

$$3\cdot 2a_3+0-2a_1+2a_1=0 \quad\Rightarrow\quad a_3=0$$

$$4\cdot 3a_4-2a_2-2\cdot 2a_2+2a_2=0 \quad\Rightarrow\quad 4\cdot 3a_4-2\cdot 2a_2=0$$

$$\Rightarrow\quad a_4=\frac{1}{3}a_2=-\frac{1}{3}a_0$$

$$5\cdot 4a_5-3\cdot 2a_3-2\cdot 3a_3+2a_3=0 \quad\Rightarrow\quad 5\cdot 4a_5=0 \quad\Rightarrow\quad a_5=0$$

$$6\cdot 5a_6 - 4\cdot 3a_4 - 2\cdot 4a_4 + 2a_4 = 0 \quad \Rightarrow \quad 30a_6 - 18a_4 = 0$$
$$\Rightarrow \quad a_6 = \frac{18}{30}a_4 = -\frac{18}{30}\frac{1}{3}a_0 = -\frac{1}{5}a_0$$

Thus, we have the solution as

$$y = \sum_{m=0}^{\infty} a_m x^m = a_0 + a_1 x - a_0 x^2 + 0x^3 - \frac{1}{3}a_0 x^4 + 0x^5 - \frac{1}{5}a_0 x^6 + \cdots$$
$$= a_1 x + a_0\left\{1 - x^2 - \frac{1}{3}x^4 - \frac{1}{5}x^6 + \cdots\right\} \equiv a_1 x - a_0 Q_1(x)$$

The solution is given as a linear sum of two elementary functions *x* and $Q_1(x)$.

It is noted that

- The ODE is the Legendre equation of order 1
- *x* is a member of Legendre polynomials, $P_1(x)$, and
- $Q_1(x)$ is a member of Legendre functions. It is an even power function.

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Convergence of power series

The power series solution of an ODE of

$$y'' + p(x)y' + q(x)y = r(x)$$

converges when all the variable coefficients are **analytic**.

A function *f* is called analytic when the function can be represented by a power series with positive radius of **convergence**.

If a partial sum s_n of a series *s* satisfies, at a point x_1, the condition of

$$\lim_{n\to\infty} s_n(x_1) = s(x_1) \quad \text{with} \quad s_n(x) = \sum_{m=0}^{n} a_m x^m$$

then, the series *s* is called convergent at $x = x_1$.

For an analytic function *f*, we can approximate the function with a partial sum s_n as accurately as we want with increasing *n*.

The convergence interval from the center is called the **radius of convergence**.
It is determined by the ratio of nearby coefficients of the series as

$$\frac{1}{R} = \lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| \quad \text{or} \quad R = 1/\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right|$$

Ex)
For a power series

$$\frac{1}{1-x} = \sum_{m=0}^{\infty} x^m = 1 + x + x^2 + \cdots$$

we have

$$\frac{1}{R} = \lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = 1 \quad \Rightarrow \quad R = 1/1 = 1$$

It means the power series converges at $|x| < 1$.

Ex)
For

$$e^x = \sum_{m=0}^{\infty} \frac{x^m}{m!} = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \cdots$$

we have

$$\frac{1}{R} = \lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\left|\frac{n!}{(n+1)!}\right| = \lim_{n\to\infty}\left|\frac{1}{n+1}\right| = 0 \quad \Rightarrow \quad R = 1/0 = \infty$$

It converges at any *x*. It is called an entire function.

Ex)
For

$$f = \sum_{m=0}^{\infty} m!\,x^m = 1 + x + 2x^2 + 6x^3 + \cdots$$

we have

$$\frac{1}{R}=\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right|=\lim_{n\to\infty}\left|\frac{(n+1)!}{n!}\right|=\lim_{n\to\infty}|n+1|=\infty \quad \Rightarrow \quad R=1/\infty=0$$

It converges only at a point, $x=0$. It is not analytic!

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Legendre's differential equation

The Legendre's differential equation arises in the Laplacian equation in the spherical coordinates, having a form of

$$\left(1-x^2\right)y''-2xy'+n(n+1)y=0$$

Or, in a standard form of linear ODE, it is

$$y''-\frac{2x}{1-x^2}y'+\frac{n(n+1)}{1-x^2}y=0$$

Since both the variable coefficients are analytic at x=0 within the circle of a radius 1, we can set the trial function with a power series as

$$y=\sum_{m=0}^{\infty}a_m x^m \quad \text{at} \quad |x|<1.$$

Substituting the trial power series function into the ODE gives

$$\left(1-x^2\right)\sum_{m=2}^{\infty}m(m-1)a_m x^{m-2}-2x\sum_{m=1}^{\infty}ma_m x^{m-1}+n(n+1)\sum_{m=0}^{\infty}a_m x^m=0$$

Collecting the terms of a like power from each summation gives

$$\sum_{m=2}^{\infty}m(m-1)a_m x^{m-2}-\sum_{m=2}^{\infty}m(m-1)a_m x^m-2\sum_{m=1}^{\infty}ma_m x^m+n(n+1)\sum_{m=0}^{\infty}a_m x^m=0$$

We can see that only the 1st term has a different power. The other three terms have the same power. So, we need to express the equation with the same power in all terms.

Since the integer m is just a summing parameter, we can adjust or replace it as

$$m-2=s \quad \text{or} \quad m=s+2$$

This adjustment makes the 1st term become

$$\sum_{m=2}^{\infty}m(m-1)a_m x^{m-2} \quad \Rightarrow \quad \sum_{s=0}^{\infty}(s+2)(s+1)a_{s+2}x^s$$

The other three terms can be adjusted to have the same summing parameter *s*, giving

$$\sum_{s=0}^{\infty}(s+2)(s+1)a_{s+2}x^s-\sum_{s=2}^{\infty}s(s-1)a_sx^s-2\sum_{s=1}^{\infty}sa_sx^s+n(n+1)\sum_{s=0}^{\infty}a_sx^s=0$$

Collecting the terms of a like power and making their coefficients vanish, we have

For $s=0$;

$$(0+2)(0+1)a_{0+2}+n(n+1)a_0=0 \quad\Rightarrow\quad 2a_2=-n(n+1)a_0$$

$$\Rightarrow\quad a_2=-\frac{n(n+1)}{2}a_0=-\frac{n(n+1)}{2!}a_0$$

For $s=1$;

$$(1+2)(1+1)a_{1+2}-2a_1+n(n+1)a_1=0 \quad\Rightarrow\quad 6a_3=-\{n(n+1)-2\}a_1$$

$$\Rightarrow\quad a_3=-\frac{(n-1)(n+2)}{6}a_1=-\frac{(n-1)(n+2)}{3!}a_1$$

For $s\ge 2$; (recurrence relation)

$$(s+2)(s+1)a_{s+2}-s(s-1)a_s-2sa_s+n(n+1)a_s=0$$

$$\Rightarrow\quad a_{s+2}=-\frac{-s(s-1)-2s+n(n+1)}{(s+2)(s+1)}a_s=-\frac{(n-s)(n+s+1)}{(s+2)(s+1)}a_s$$

It means that any coefficients can be obtained from the first two coefficients a_0 and a_1.

For an example

$$\begin{aligned}a_4=a_{2+2}&=-\frac{(n-2)(n+2+1)}{(2+2)(2+1)}a_2=-\frac{(n-2)(n+3)}{4\cdot 3}\left\{-\frac{n(n+1)}{2!}\right\}a_0\\&=+\frac{(n-2)n(n+1)(n+3)}{4!}a_0\end{aligned}$$

and

$$a_5=a_{3+2}=-\frac{(n-3)(n+3+1)}{(3+2)(3+1)}a_3=-\frac{(n-3)(n+4)}{5\cdot 4}\left\{-\frac{(n-1)(n+2)}{3!}\right\}a_1$$

$$= -\frac{(n-3)(n-1)(n+2)(n+4)}{5!}a_1$$

Similarly, we have

$$a_6 = a_{4+2} = -\frac{(n-4)(n+4+1)}{(4+2)(4+1)}a_4 = -\frac{(n-4)(n+5)}{6\cdot 5}a_4$$

$$= -\frac{(n-4)(n+5)}{6\cdot 5}\frac{(n-2)n(n+1)(n+3)}{4!}a_0$$

$$= -\frac{(n-4)(n-2)n(n+1)(n+3)(n+5)}{6!}a_0$$

and

$$a_7 = +\frac{(n-5)(n-3)(n-1)(n+2)(n+4)(n+6)}{7!}a_1$$

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It gives us two solutions. One is proportional to a_0 and the other proportional to a_1,

$$y = a_0 y_1(x) + a_1 y_2(x).$$

$$y_1(x) = \frac{a_0}{a_0}x^0 + \frac{a_2}{a_0}x^2 + \frac{a_4}{a_0}x^4 + \cdots$$

$$= 1 - \frac{n(n+1)}{2!}x^2 + \frac{(n-2)n(n+1)(n+3)}{4!}x^4 - + \cdots$$

$$y_2(x) = \frac{a_1}{a_1}x^1 + \frac{a_3}{a_1}x^2 + \frac{a_5}{a_1}x^4 + \cdots$$

$$= x - \frac{(n-1)(n+2)}{3!}x^3 + \frac{(n-3)(n-1)(n+2)(n+4)}{5!}x^5 - + \cdots$$

When the series is made infinitely, it is valid only at $-1 < x < 1$.
However, when the series is terminated at some order, it becomes valid for all *x*.
The termination happens when *n* is an integer.

From the <u>recurrence relation</u> of

$$a_{s+2} = -\frac{(n-s)(n+s+1)}{(s+2)(s+1)}a_s$$

we can see that when *n* is a nonnegative integer, the coefficient a_{n+2} vanishes, since

$$a_{n+2} = -\frac{(n-n)(n+n+1)}{(n+2)(n+1)} a_n = 0$$

Thus, we have

$$a_{n+2} = a_{n+4} = a_{n+6} \cdots = 0$$

The highest order of the power series becomes the order *n*. It gives the Legendre polynomial of order *n*. It is just one elementary solution of the Legendre's equation.

By assigning the coefficient a_0 or a_1 to have the highest order coefficient a_n as

$$a_n = \frac{(2n)!}{2^n (n!)^2}$$

we have the Legendre polynomial of order *n* as

$$P_n(x) = \sum_{m=0}^{M} (-1)^m \frac{(2n-2m)!}{2^n m!(n-m)!(n-2m)!} x^{n-2m}$$

Where, the integer *M* is defined

$$M = \frac{n}{2} \text{ for even } n$$

$$M = \frac{n-1}{2} \text{ for odd } n$$

We can check that, for every *n*, it is normalized as

$$P_n(x=1) = 1.$$

It is noted that the Legendre polynomials are orthogonal to each other in the interval of $-1 \le x \le 1$.

In $y = a_0 y_1(x) + a_1 y_2(x)$, we can prove that one is a terminated polynomial but the other is not.

In detail, for even *n*, $y_1(x)$ becomes terminated to have the polynomial of order *n*.

Whilst, $y_2(x)$ is a polynomial of infinite order.

However, for odd *n*, $y_2(x)$ becomes terminated to have the polynomial of order *n*.

Whilst, $y_1(x)$ becomes a polynomial of infinite order instead.

The terminated polynomials are the **Legendre polynomials**.
They are orthogonal to each other!
The others are called **Legendre functions**.